Block practice · More diet, digestion and teeth

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Explain how food moves by peristalsis and why fibre helps

Explain causes of obesity, its health risks and dietary advice

Calculate BMI and place it in a weight category

Apply digestive enzyme action to an unfamiliar situation

OLevels.com SCHOOL

Practice workbook · Explain how food moves by peristalsis and why fibre helps

Movement of the bolus along oesophagus and gut by muscle waves, and the role of dietary fibre.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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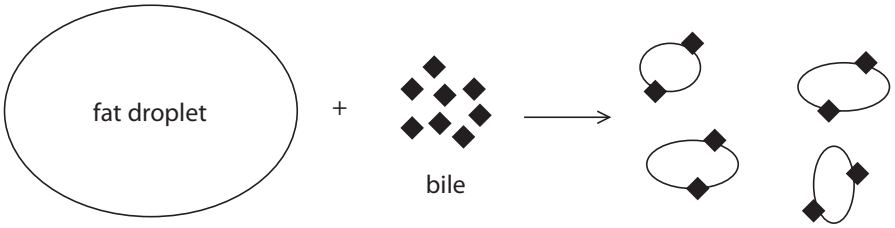
Type ID: HBI-PERISTALSIS-FIBRE

1. Question 3 · part c · 5 marks

January 2023 Paper 1

Worked-example retry · Attempt part c only.

3 (a) The diagram represents the effect of bile on a large fat droplet.



(i) Where is bile produced?

(1)

- ☒ **A** gall bladder
- ☒ **B** liver
- ☒ **C** small intestine
- ☒ **D** stomach

(ii) Describe the role of bile in the digestion of fat.

(3)

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1. Question 3 · part c · 5 marks

January 2023 Paper 1

Worked-example retry · Attempt part c only.

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(b) Three food tests are used on a solution of food.

The table shows the reagent used and the colour of the food solution after each test.

Reagent used	Colour of the food solution after the test
iodine solution	blue/black
biuret solution	purple
Benedict's solution	blue

(i) Describe what the food tests show about the dietary components in the food.

(3)

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(ii) Saliva is added to a new sample of the food solution and the mixture is kept at 37 °C for 10 minutes.

The food tests are then repeated and the solution turns brick red when tested with Benedict's reagent.

Explain the effect of saliva on the food solution.

(3)

1. Question 3 · part c · 5 marks

January 2023 Paper 1

Worked-example retry · Attempt part c only.

(c) Food is digested as it moves along the alimentary canal.

This process is helped by a diet containing fibre.

Explain how food is moved along the alimentary canal and how fibre helps this process.

(5)

(Total for Question 3 = 15 marks)

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1. Question 3 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 1 · part c · 3 marks

November 2020 Paper 2R

Further practice · Attempt part c only.

1 Food enters the alimentary canal through the mouth.

All undigested food leaves the body through the anus.

(a) Name three structures the food passes through between the mouth and the anus.

(3)

1 .....

2 .....

3 .....

(b) Some of the food is digested by enzymes as it passes through the alimentary canal.

(i) Name an enzyme that digests protein.

(1)

.....

(ii) Describe the processes that occur in the mouth to digest food.

(4)

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OLevels.com SCHOOL

2. Question 1 · part c · 3 marks

November 2020 Paper 2R

Further practice · Attempt part c only.

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(c) Fibre in food cannot be digested.

Explain the importance of fibre in the diet.

(3)

(Total for Question 1 = 11 marks)

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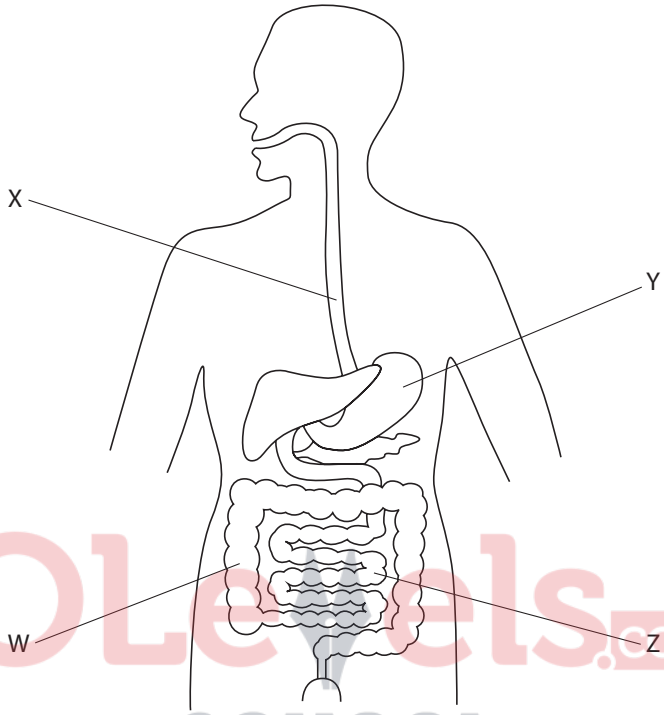
Working and answer

3. Question 2 · part a.ii · 4 marks

June 2022 Paper 1R

Further practice · Attempt part a.ii only.

2 (a) The diagram shows the human digestive system.



(i) Name the structures labelled W, X, Y and Z.

(4)

W .....

X .....

Y .....

Z .....

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P 7 0 9 4 9 A 0 6 2 4

3. Question 2 · part a.ii · 4 marks

June 2022 Paper 1R

Further practice · Attempt part a.ii only.

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(ii) The box gives words used to describe how food is moved through structure X towards structure Y.

relax	coordination	contract	liquid
peristalsis	muscles	enzymes	bolus

Use words from the box to complete the passage.

(4)

Food forms a ..... at the back of the mouth.

The food is swallowed and passes into structure X. The walls of structure X contain .....

which ..... to push the food towards structure Y.

This process is known as ..... .

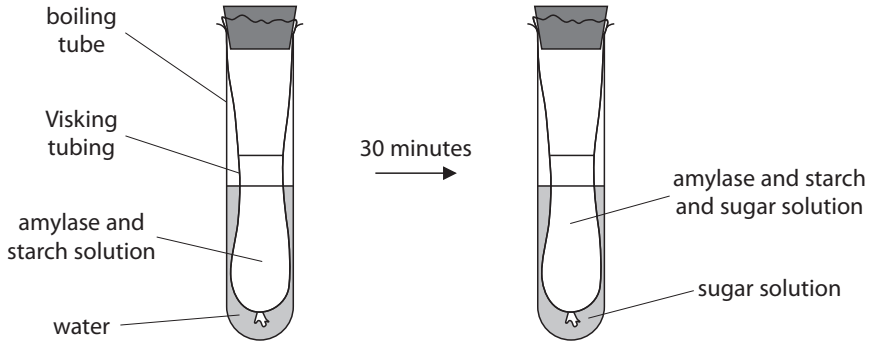
OLevels.COM — SCHOOL —

3. Question 2 · part a.ii · 4 marks

June 2022 Paper 1R

Further practice · Attempt part a.ii only.

(b) The diagram shows some apparatus used to investigate the effect of an enzyme called amylase on starch.



(i) Name one area in the digestion system where amylase is produced.

(1)

(ii) Name the structure in the digestive system that the Visking tubing represents.

(1)

(iii) Explain the results of the investigation after 30 minutes.

(2)

(iv) Name the chemical used to test for sugar in the water surrounding the Visking tubing after 30 minutes.

(1)

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3. Question 2 · part a.ii · 4 marks · Working and answer

Working and answer

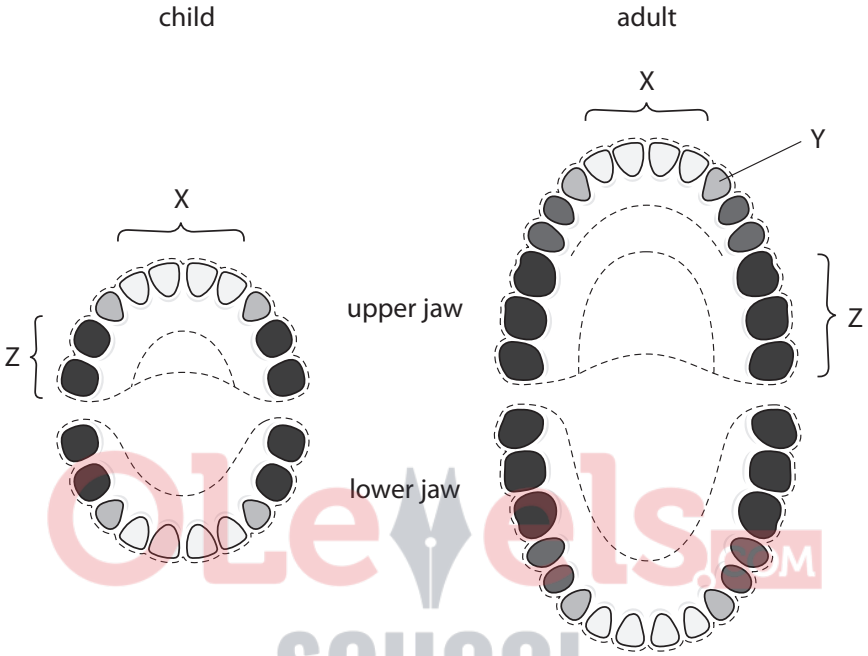
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4. Question 1 · part c.ii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part c.ii only.

1 (a) The diagrams show the arrangement of teeth in a young child and the arrangement of teeth in an adult.



(Source: https://www.medicalnewstoday.com/articles/326754#number-of-teeth)

(i) Name the types of teeth shown by X and Z.

(2)

X .....

Z .....

(ii) State the function of the tooth labelled Y.

(1)

.....

.....

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4. Question 1 · part c.ii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part c.ii only.

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(b) Digestion of food begins in the mouth.

Use the most appropriate words or numbers to complete the passage about digestion of food in the mouth.

(4)

Saliva in the mouth contains an enzyme called ..... .

This enzyme speeds up the digestion of starch to produce

..... .

The enzyme in saliva works best at pH ..... .

If the pH in the mouth was too low or too high, the digestion of carbohydrates

would ..... .

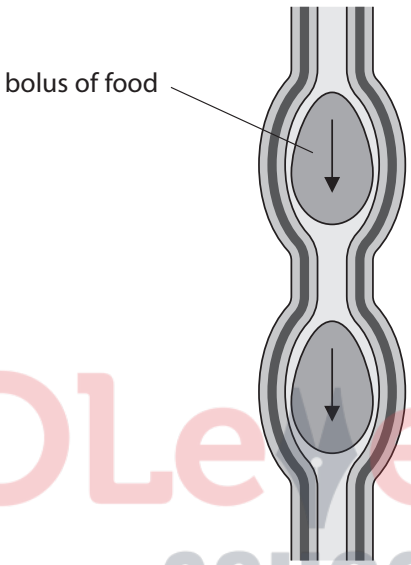
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4. Question 1 · part c.ii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part c.ii only.

(c) The diagram shows the part of the digestive system that transports food from the mouth to the stomach.



(Source: https://www.motifolio.com/1031265.html)

(i) Name the part of the digestive system that transports food from the mouth to the stomach.

(1)

.....

(ii) Describe how the action of this part of the digestive system transports food from the mouth to the stomach.

(2)

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(Total for Question 1 = 10 marks)

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4. Question 1 · part c.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Explain causes of obesity, its health risks and dietary advice

Energy imbalance causing obesity, its links to diabetes and heart disease, and diet changes to reduce mass.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: HBI-OBESITY-CAUSES-RISKS

1. Question 4 · part a.i · 3 marks

June 2021 Paper 2

Worked-example retry · Attempt part a.i only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Obesity and diabetes have been linked to a 20% increase in the number of leg and foot amputations.

Data from Diabetes UK shows that there were 26 378 lower limb amputations related to diabetes from the period 2014 to 2017. This is an increase from the period 2010 to 2013 when there were 22 092 amputations related to diabetes.

Minor lower limb amputations are below the ankle and major lower limb amputations are below the knee. Minor lower limb amputations increased by 26.5% and major lower limb amputations increased by 4.0%.

In 2019 in the United Kingdom 3.8 million people had been diagnosed with diabetes and 90% of these people had Type 2 diabetes.

(a) (i) Describe the causes of obesity.

(3)

OLevels.COM SCHOOL

(ii) Give two health risks, other than diabetes, associated with obesity.

(2)

1 .....

.....

2 .....

.....

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1. Question 4 · part a.i · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 5 · part b.ii · 2 marks

June 2019 Paper 1

Further practice · Attempt part b.ii only.

5 A teacher wants to calculate the body mass index (BMI) of a number of students.

(a) The teacher measures the height of students from four age groups.

There are two male and two female students in each age group.

The table shows the data collected.

Age in years	Height of student in cm				Mean height of students in cm
	Male		Female		
12	146.2	148.9	142.8	144.2	145.5
14	165.7	166.4	161.9	164.0	164.5
16	175.9	178.5	166.3	167.6	
18	180.9	181.3	171.2	174.3	176.9

(i) Calculate the mean height of the students aged 16.

Give your answer to one decimal place.

(2)

OLevels.com SCHOOL

mean height = ... cm

(ii) State two conclusions that the teacher could make from the data in the table.

(2)

1 .....

.....

2 .....

.....

(iii) Give two ways the teacher could improve his investigation.

(2)

1 .....

.....

2 .....

.....

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2. Question 5 · part b.ii · 2 marks

June 2019 Paper 1

Further practice · Attempt part b.ii only.

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(b) The table shows data for two different students.

Student	Height in m	Mass in kg	BMI
X	1.69	62	21.7
Y	1.46	71	

(i) BMI is calculated using the equation

$$\text{BMI} = \frac{\text{mass}}{\text{height}^2}$$

Calculate the BMI of student Y.
[mass measured in kg, height measured in m]

(2)

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BMI of student Y = ...

(ii) BMI can be used to assess whether a person has a healthy mass for their height.

BMI	Classification
below 18.5	underweight
18.5 – 24.9	normal weight
25 – 29.9	overweight
30 – 40	obese
above 40	morbidly obese

Using information from the table, explain how the BMI of student Y could affect her health.

(2)

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(Total for Question 5 = 10 marks)

2. Question 5 · part b.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 1 · part b.ii · 1 mark

January 2020 Paper 1

Further practice · Attempt part b.ii only.

1 (a) Food packaging gives information about the food it contains.

The photograph shows some of the information given on a package containing yogurt.



(Source: © studiomode / Alamy Stock Photo)

The instructions state that the yogurt should be stored in a fridge at 4°C.

Why should this instruction be followed?

(1)

- ☒ **A** to make the yogurt taste better
- ☒ **B** to kill all bacteria in the yogurt
- ☒ **C** to slow down the growth of bacteria in the yogurt
- ☒ **D** to improve the appearance of the yogurt

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3. Question 1 · part b.ii · 1 mark

January 2020 Paper 1

Further practice · Attempt part b.ii only.

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(b) The diagram shows the nutritional label on a package containing yogurt.

Nutrition Facts

Serving Size
1 container (150g)
Energy in kJ 480
Energy from Fat 0

Amount/Serving	% Daily Value*	Amount/Serving	% Daily Value*
Total Fat	a%	Sodium	2%
Saturated Fat	0%	Total Carb.	6%
Trans Fat		Sugars	
Cholesterol	1%	Protein	22%

Vitamin A 4% • Calcium 10% • Vitamin D 20%

Not a significant source of dietary fibre, vitamin C and iron.

(i) State which nutrient on the label is required for growth and repair of body tissue.

(1)

.....

(ii) Why is this yogurt suitable for a person who wants to reduce their body mass?

(1)

☒ **A** it contains vitamins C and D

☒ **B** it contains less than 5 mg of cholesterol

☒ **C** it does not contain energy from carbohydrates

☒ **D** it does not contain energy from fat

(c) Yogurt contains reducing sugars.

Describe a method to test for reducing sugars.

(3)

.....

.....

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.....

(Total for Question 1 = 6 marks)

3. Question 1 · part b.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 4 · part a.ii · 2 marks

June 2021 Paper 2

Further practice · Attempt part a.ii only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Obesity and diabetes have been linked to a 20% increase in the number of leg and foot amputations.

Data from Diabetes UK shows that there were 26 378 lower limb amputations related to diabetes from the period 2014 to 2017. This is an increase from the period 2010 to 2013 when there were 22 092 amputations related to diabetes.

Minor lower limb amputations are below the ankle and major lower limb amputations are below the knee. Minor lower limb amputations increased by 26.5% and major lower limb amputations increased by 4.0%.

In 2019 in the United Kingdom 3.8 million people had been diagnosed with diabetes and 90% of these people had Type 2 diabetes.

(a) (i) Describe the causes of obesity.

(3)

(ii) Give two health risks, other than diabetes, associated with obesity.

(2)

1 .....

2 .....

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OLevels.com SCHOOL

4. Question 4 · part a.ii · 2 marks

June 2021 Paper 2

Further practice · Attempt part a.ii only.

(b) Diabetes is caused as a result of insulin not being secreted or not working correctly.

(i) Which organ secretes insulin?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

(ii) Insulin stimulates the conversion of glucose into glycogen.

In which organ does this occur?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

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(c) (i) Calculate the percentage increase in the number of lower limb amputations between the period 2010 to 2013 and the period 2014 to 2017.

(3)

percentage increase = ...%

(ii) Calculate the number of people in 2019 who had been diagnosed with Type 2 diabetes in the United Kingdom.

(1)

number of people = ... million

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4. Question 4 · part a.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 3 · part e · 1 mark

November 2021 Paper 2

Further practice · Attempt part e only.

3 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Sweating is normal when we are hot. It is part of our natural cooling mechanism. If you sweat visibly when you do not need to cool down, it can be a problem.

The sweating process is controlled by the brain, which sends signals down the nerves to the sweat glands.

Sweating is crucial to survival. We sweat to control body temperature and the sweat glands in our armpits can produce several litres of sweat in 24 hours. How much we sweat varies hugely from individual to individual and even from day to day.

Hyperhidrosis occurs when some areas of the body start sweating excessively. This affects about 1% of the population. There are two main types. The most common type is primary focal hyperhidrosis, which affects many parts of the body. There is no known cause, but it usually begins in childhood and often runs in families. Another type is secondary hyperhidrosis, which often doesn't begin until after the teenage years, and usually has an underlying medical cause. These causes can include diabetes, menopause or chronic infection. It can also be a result of eating certain foods. Stress can play a major role too.

(a) Which part of the brain controls sweating (line 4)?

(1)

- ☒ **A** cerebellum
- ☒ **B** cerebral hemispheres
- ☒ **C** hypothalamus
- ☒ **D** pituitary gland

(b) Which type of nerve sends signals to the sweat glands (lines 4 and 5)?

(1)

- ☒ **A** connector
- ☒ **B** motor
- ☒ **C** relay
- ☒ **D** sensory

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5. Question 3 · part e · 1 mark

November 2021 Paper 2

Further practice · Attempt part e only.

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(c) Describe how sweating controls body temperature (lines 6 and 7).

(3)

(d) Explain the evidence in the passage that indicates primary focal hyperhidrosis is a genetic condition.

(3)

OLevels.com SCHOOL

(e) Secondary hyperhidrosis can be caused by diabetes.

Give one risk factor for diabetes (line 16).

(1)

5. Question 3 · part e · 1 mark

November 2021 Paper 2

Further practice · Attempt part e only.

(f) The passage states that how much we sweat varies from individual to individual and from day to day.

Explain why this is the case in people without hyperhidrosis (lines 8 and 9).

(5)

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(Total for Question 3 = 14 marks)

5. Question 3 · part e · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 3 · part c.ii · 2 marks

June 2022 Paper 1R

Further practice · Attempt part c.ii only.

3 (a) In England, 3 319 266 people have diabetes.

(i) Give this value to 3 significant figures.

(1)

people with diabetes = ...

(ii) In the rest of the United Kingdom, 600 239 people have diabetes.

Give this value in standard form.

(2)

people with diabetes = ...

(iii) Diabetes can be caused by the inability of an organ to produce insulin.

Name the organ that produces insulin.

(1)

.....

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6. Question 3 · part c.ii · 2 marks

June 2022 Paper 1R

Further practice · Attempt part c.ii only.

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(b) The body mass index (BMI) is a measure that uses height and body mass to work out if the body mass is healthy.

The table shows the BMI values and the categories into which a person can be placed.

BMI	BMI category
less than 18.5	underweight
18.5 to 24.9	healthy weight
25.0 to 29.9	overweight
30.0 to 39.9	obese
40.0 and over	morbidly obese

OLevels.COM SCHOOL

(i) Calculate the BMI of a person with a body mass of 120 kg and a height of 1.95 m.

Use the equation

$$\text{BMI} = \frac{\text{body mass}}{\text{height}^2}$$

(2)

BMI = ...

(ii) Determine the BMI category that this person belongs to.

(1)

.....

.....

.....

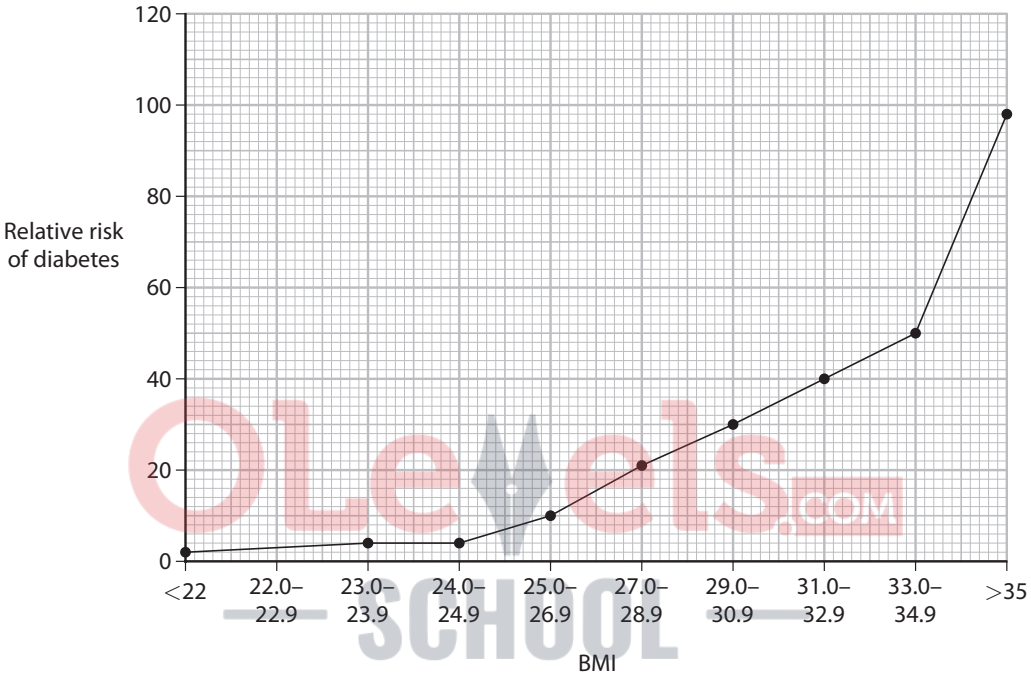
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6. Question 3 · part c.ii · 2 marks

June 2022 Paper 1R

Further practice · Attempt part c.ii only.

(c) The graph shows the relative risk of developing a form of diabetes called type 2 for different BMI values.



(Source: adapted from Colditz GA, et al., *Ann Intern Med.*, 1995; 122:481–486)

(i) Describe the trend shown by the graph.

(2)

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6. Question 3 · part c.ii · 2 marks

June 2022 Paper 1R

Further practice · Attempt part c.ii only.

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(ii) Give some dietary advice that could help an obese person lower their BMI.

(2)

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(Total for Question 3 = 11 marks)

Working and answer

OLevels.com SCHOOL

Practice workbook · Calculate BMI and place it in a weight category

Body mass index from mass and height and reading its category from a table.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: HBI-BMI-CALCULATE-CLASSIFY

1. Question 3 · part b.i · 2 marks

June 2022 Paper 1R

Worked-example retry · Attempt part b.i only.

3 (a) In England, 3 319 266 people have diabetes.

(i) Give this value to 3 significant figures.

(1)

people with diabetes = ...

(ii) In the rest of the United Kingdom, 600 239 people have diabetes.

Give this value in standard form.

(2)

people with diabetes = ...

(iii) Diabetes can be caused by the inability of an organ to produce insulin.

Name the organ that produces insulin.

(1)

.....

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1. Question 3 · part b.i · 2 marks

June 2022 Paper 1R

Worked-example retry · Attempt part b.i only.

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(b) The body mass index (BMI) is a measure that uses height and body mass to work out if the body mass is healthy.

The table shows the BMI values and the categories into which a person can be placed.

BMI	BMI category
less than 18.5	underweight
18.5 to 24.9	healthy weight
25.0 to 29.9	overweight
30.0 to 39.9	obese
40.0 and over	morbidly obese

OLevels.com SCHOOL

(i) Calculate the BMI of a person with a body mass of 120 kg and a height of 1.95 m.

Use the equation

$$\text{BMI} = \frac{\text{body mass}}{\text{height}^2}$$

(2)

BMI = ...

(ii) Determine the BMI category that this person belongs to.

(1)

.....

.....

.....

.....

1. Question 3 · part b.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 5 · part b.i · 2 marks

June 2019 Paper 1

Further practice · Attempt part b.i only.

5 A teacher wants to calculate the body mass index (BMI) of a number of students.

(a) The teacher measures the height of students from four age groups.

There are two male and two female students in each age group.

The table shows the data collected.

Age in years	Height of student in cm				Mean height of students in cm
	Male		Female		
12	146.2	148.9	142.8	144.2	145.5
14	165.7	166.4	161.9	164.0	164.5
16	175.9	178.5	166.3	167.6	
18	180.9	181.3	171.2	174.3	176.9

(i) Calculate the mean height of the students aged 16.

Give your answer to one decimal place.

(2)

OLevels.com SCHOOL

mean height = ... cm

(ii) State two conclusions that the teacher could make from the data in the table.

(2)

1 .....

.....

2 .....

.....

(iii) Give two ways the teacher could improve his investigation.

(2)

1 .....

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2. Question 5 · part b.i · 2 marks

June 2019 Paper 1

Further practice · Attempt part b.i only.

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(b) The table shows data for two different students.

Student	Height in m	Mass in kg	BMI
X	1.69	62	21.7
Y	1.46	71	

(i) BMI is calculated using the equation

$$\text{BMI} = \frac{\text{mass}}{\text{height}^2}$$

Calculate the BMI of student Y.
[mass measured in kg, height measured in m]

(2)

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BMI of student Y = ...

(ii) BMI can be used to assess whether a person has a healthy mass for their height.

BMI	Classification
below 18.5	underweight
18.5 – 24.9	normal weight
25 – 29.9	overweight
30 – 40	obese
above 40	morbidly obese

Using information from the table, explain how the BMI of student Y could affect her health.

(2)

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(Total for Question 5 = 10 marks)

2. Question 5 · part b.i · 2 marks · Working and answer

Working and answer

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3. Question 3 · part b.ii · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.ii only.

3 (a) In England, 3 319 266 people have diabetes.

(i) Give this value to 3 significant figures.

(1)

people with diabetes = ...

(ii) In the rest of the United Kingdom, 600 239 people have diabetes.

Give this value in standard form.

(2)

people with diabetes = ...

(iii) Diabetes can be caused by the inability of an organ to produce insulin.

Name the organ that produces insulin.

(1)

.....

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3. Question 3 · part b.ii · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.ii only.

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(b) The body mass index (BMI) is a measure that uses height and body mass to work out if the body mass is healthy.

The table shows the BMI values and the categories into which a person can be placed.

BMI	BMI category
less than 18.5	underweight
18.5 to 24.9	healthy weight
25.0 to 29.9	overweight
30.0 to 39.9	obese
40.0 and over	morbidly obese

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(i) Calculate the BMI of a person with a body mass of 120 kg and a height of 1.95 m.

Use the equation

$$\text{BMI} = \frac{\text{body mass}}{\text{height}^2}$$

(2)

BMI = ...

(ii) Determine the BMI category that this person belongs to.

(1)

.....

.....

.....

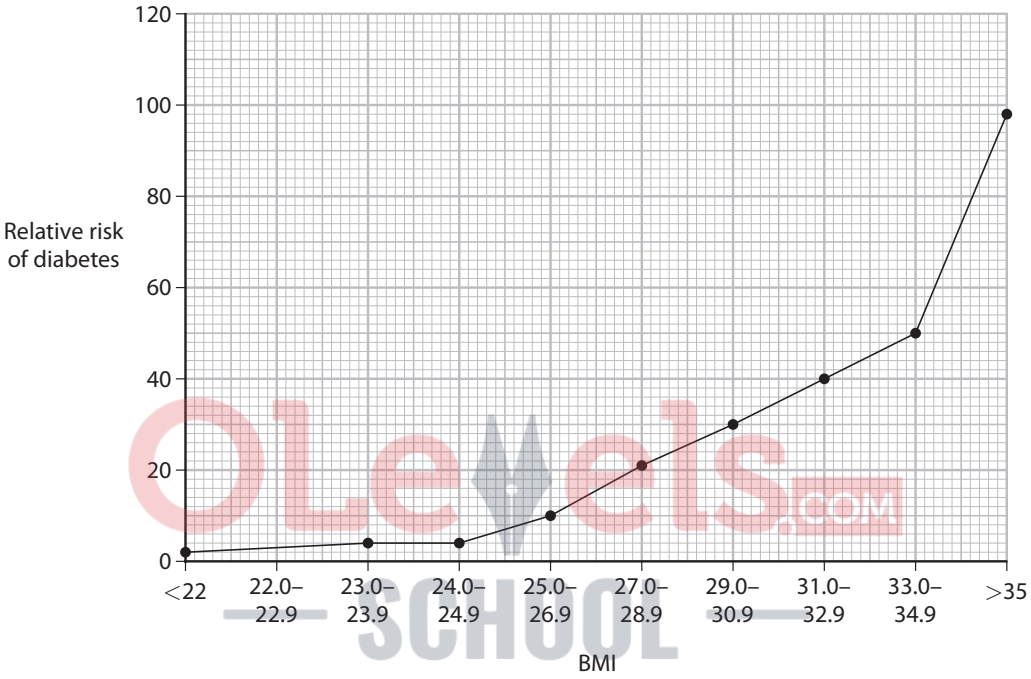
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3. Question 3 · part b.ii · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.ii only.

(c) The graph shows the relative risk of developing a form of diabetes called type 2 for different BMI values.



(Source: adapted from Colditz GA, et al., *Ann Intern Med.*, 1995; 122:481–486)

(i) Describe the trend shown by the graph.

(2)

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3. Question 3 · part b.ii · 1 mark · Working and answer

Working and answer

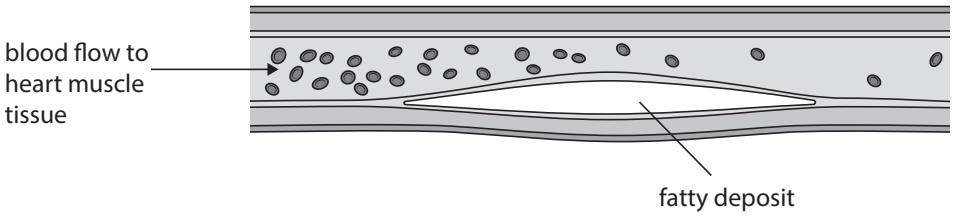
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4. Question 7 · part b.ii · 3 marks

June 2024 Paper 2

Further practice · Attempt part b.ii only.

7 (a) The diagram shows a fatty deposit in an artery carrying blood to the heart muscle.



(Source: adapted from https://www.heartandstroke.ca/heart-disease/conditions/atherosclerosis)

Explain how a build-up of a fatty deposit in an artery carrying blood to the heart muscle can cause heart disease.

(4)

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4. Question 7 · part b.ii · 3 marks

June 2024 Paper 2

Further practice · Attempt part b.ii only.

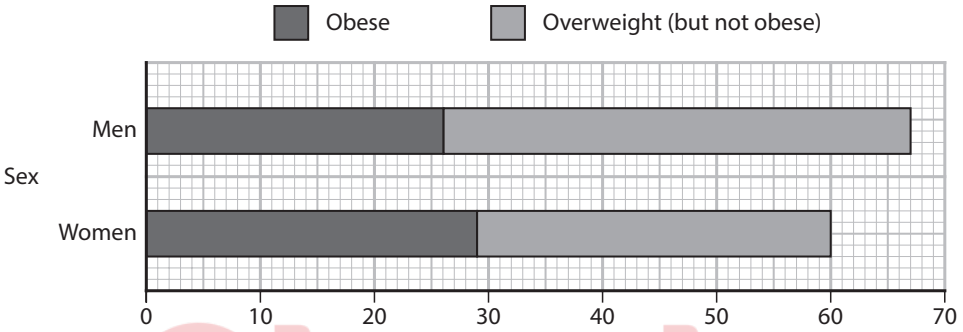
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(b) People with obesity are more at risk of heart disease than those who are overweight.

The chart shows the percentage of men and women in the United Kingdom who were either overweight or obese in 2018.



(Source: https://digital.nhs.uk/data-and-information/publications/statistical/health-survey-for-england/2018/summary)

(i) In 2018 there were 66.5 million people living in the UK. Of these people, 33 million were men.

Calculate how many more women than men were classified as obese.

(3)

extra number of women = ...

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P 7 4 6 3 0 A 0 1 9 2 4

Turn over ▶

4. Question 7 · part b.ii · 3 marks

June 2024 Paper 2

Further practice · Attempt part b.ii only.

(ii) Describe a calculation that could be used to determine if a person is obese.

(3)

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4. Question 7 · part b.ii · 3 marks

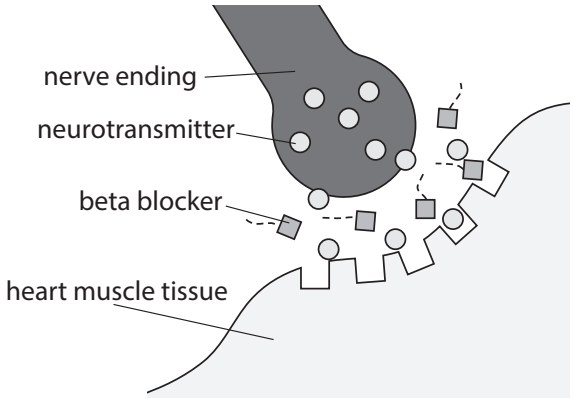
June 2024 Paper 2

Further practice · Attempt part b.ii only.

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(c) The diagram shows a junction between a neurone and heart muscle.



(Source: Adapted from: https://study.com/academy/lesson/what-are-beta-blockers-uses-types-examples-side-effects.html)

Use the diagram to explain how beta blockers can reduce the risk of heart disease.

(3)

(Total for Question 7 = 13 marks)

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4. Question 7 · part b.ii · 3 marks · Working and answer

Working and answer

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5. Question 4 · part b.i · 2 marks

November 2024 Paper 1

Further practice · Attempt part b.i only.

4 Carbohydrates are large molecules found in many body cells.

(a) Glycogen is a carbohydrate.

(i) Describe how the liver produces glycogen.

(2)

(ii) Name the hormone that stimulates the liver to produce glycogen.

(1)

(iii) Name a hormone that stimulates the liver to increase blood sugar levels.

(1)

(b) Consuming a diet high in carbohydrates can increase the BMI of a person.

The table gives information on the BMI categories of individuals.

BMI	Category
less than 18.5	underweight
18.5 to 24.9	healthy weight
25 to 29.9	overweight
30 to 34.9	obese
35 to 39.9	severely obese
more than 39.9	morbidly obese

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5. Question 4 · part b.i · 2 marks

November 2024 Paper 1

Further practice · Attempt part b.i only.

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(i) A person has a height of 1.69 m and a mass of 67.0 kg.

Calculate the BMI of this person.

(2)

BMI .....

(ii) Give the BMI category of this person.

(1)

.....

(c) Obesity is a risk factor for heart disease.

Explain how obesity can cause heart disease.

(2)

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(Total for Question 4 = 9 marks)

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Working and answer

6. Question 4 · part b.ii · 1 mark

November 2024 Paper 1

Further practice · Attempt part b.ii only.

4 Carbohydrates are large molecules found in many body cells.

(a) Glycogen is a carbohydrate.

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(2)

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(1)

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6. Question 4 · part b.ii · 1 mark

November 2024 Paper 1

Further practice · Attempt part b.ii only.

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(i) A person has a height of 1.69 m and a mass of 67.0 kg.

Calculate the BMI of this person.

(2)

BMI .....

(ii) Give the BMI category of this person.

(1)

.....

(c) Obesity is a risk factor for heart disease.

Explain how obesity can cause heart disease.

(2)

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(Total for Question 4 = 9 marks)

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Working and answer

Practice workbook · Apply digestive enzyme action to an unfamiliar situation

Novel contexts where digestion, or its absence, changes what is absorbed: oral hormones, PKU, chymosin, lactase loss.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-ENZYME-APPLIED-CONTEXT

1. Question 9 · part a.ii · 3 marks

January 2020 Paper 2

Worked-example retry · Attempt part a.ii only.

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9 (a) Phenylketonuria (PKU) is a serious genetic disorder.

People with PKU are unable to break down the amino acid, phenylalanine.

Phenylalanine builds up in the blood and in the brain causing brain damage.

(i) Describe how levels of phenylalanine are reduced in the body of a healthy person.

(3)

(ii) Explain why people with PKU are advised to follow a low-protein diet.

(3)

(iii) Explain the impact of a low-protein diet on protein synthesis in the cytoplasm of body cells.

(3)

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1. Question 9 · part a.ii · 3 marks · Working and answer

Working and answer

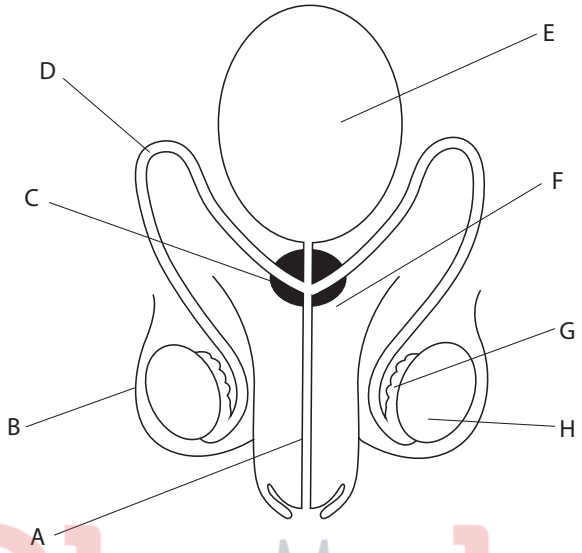
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2. Question 7 · part c.i · 2 marks

June 2019 Paper 2R

Further practice · Attempt part c.i only.

7 (a) The diagram shows the male reproductive system.



The table gives descriptions of structures in the male reproductive system.

Complete the table by giving a letter from the diagram to match each description.

Each letter may be used once, more than once or not at all.

(4)

Description of the structure	Letter
stores sperm	
produces sperm	
produces fluid to help sperm move	
contains cells dividing by meiosis	

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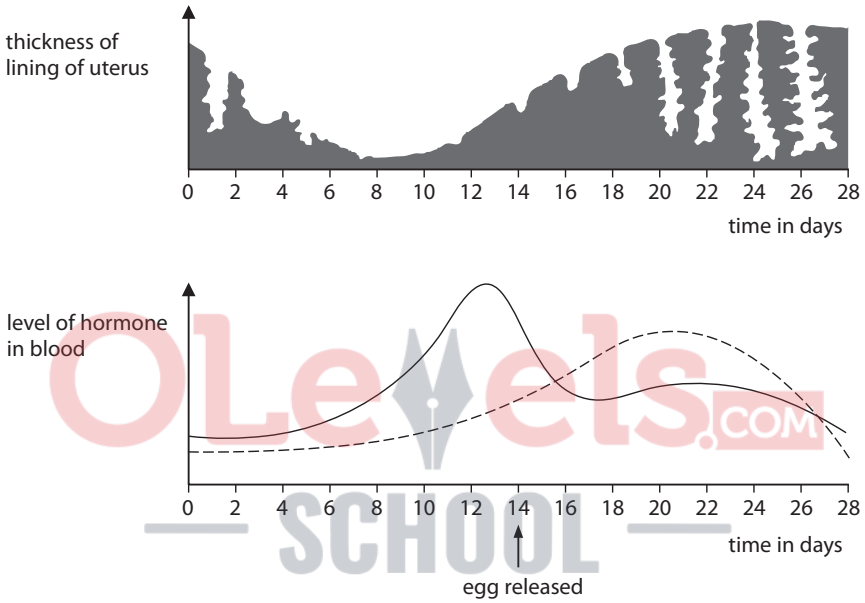
2. Question 7 · part c.i · 2 marks

June 2019 Paper 2R

Further practice · Attempt part c.i only.

(b) Changes occur in the female body during the menstrual cycle.

The graphs show changes in the lining of the uterus and the levels of two hormones that control these changes.



(i) Use information from the graphs to explain the cause of growth in the lining of the uterus after day 12.

(2)

(ii) Explain why fertilisation is less likely to occur from intercourse between days 8 and 12.

(2)

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2. Question 7 · part c.i · 2 marks

June 2019 Paper 2R

Further practice · Attempt part c.i only.

(c) A synthetic oestrogen called oestradiol is used to make contraceptive pills. Natural oestrogen, which is a lipid molecule, could be used but is less effective than oestradiol when taken as a pill.

(i) Suggest why natural oestrogen is not as effective as oestradiol when taken as a pill.
(2)

(ii) Give a reason why contraceptive pills only need to be taken for 21 days out of the 28-day cycle.
(1)

(Total for Question 7 = 11 marks)

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Working and answer

3. Question 7 · part a.i · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.i only.

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7 Chymosin is an enzyme found in young humans, but not in adults. It converts soluble proteins in milk into solid proteins.

(a) (i) Suggest the advantages to a young human of having chymosin in their alimentary canal.

(2)

(ii) Explain why chymosin is only needed in young humans.

(2)

(iii) Describe a test to show that the solid formed by the action of chymosin on milk is a protein.

(3)

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3. Question 7 · part a.i · 2 marks · Working and answer

Working and answer

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4. Question 5 · part b · 1 mark

June 2022 Paper 2R

Further practice · Attempt part b only.

5 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Milk contains a carbohydrate called lactose. Lactose is not absorbed by the epithelium of the ileum or duodenum.

Lactose is a disaccharide which is broken down into glucose and galactose by the enzyme lactase. Babies produce large amounts of lactase but in some people the gene for producing lactase is switched off as they get older. When this gene is switched off people then become lactose-intolerant which means they cannot digest any lactose that is taken into the gut.

The number of people with lactose intolerance differs across the world. People in some countries are often tolerant because their ancestors have evolved tolerance because of their diet.

In a lactose intolerant person, any lactose taken into the gut will collect and ferment, forming methane and carbon dioxide plus other soluble products. These other soluble products lower the water potential in the gut.

Symptoms of lactose intolerance include diarrhoea, calcium deficiency and vitamin D deficiency.

(a) Suggest what is meant by the term disaccharide. (lines 3 to 4) **(2)**

(b) Give a reason why lactose is not absorbed by the epithelium. (lines 1 to 2) **(1)**

(c) (i) Explain what is meant by a gene being switched off. (lines 5 to 6) **(3)**

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4. Question 5 · part b · 1 mark · Working and answer

Working and answer

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5. Question 5 · part c.ii · 2 marks

June 2022 Paper 2R

Further practice · Attempt part c.ii only.

5 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Milk contains a carbohydrate called lactose. Lactose is not absorbed by the epithelium of the ileum or duodenum.

Lactose is a disaccharide which is broken down into glucose and galactose by the enzyme lactase. Babies produce large amounts of lactase but in some people the gene for producing lactase is switched off as they get older. When this gene is switched off people then become lactose-intolerant which means they cannot digest any lactose that is taken into the gut.

The number of people with lactose intolerance differs across the world. People in some countries are often tolerant because their ancestors have evolved tolerance because of their diet.

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Symptoms of lactose intolerance include diarrhoea, calcium deficiency and vitamin D deficiency.

(a) Suggest what is meant by the term disaccharide. (lines 3 to 4) **(2)**

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5. Question 5 · part c.ii · 2 marks

June 2022 Paper 2R

Further practice · Attempt part c.ii only.

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(ii) Describe the effects on a person when the gene for producing lactase is switched off. (lines 3 to 7)

(2)

(d) Suggest how people in some parts of the world have evolved tolerance because of their diet. (lines 8 to 10)

(2)

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(e) Explain how the symptoms of lactose intolerance are caused. (lines 11 to 15)

(4)

(Total for Question 5 = 14 marks)

5. Question 5 · part c.ii · 2 marks · Working and answer

Working and answer

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6. Question 7 · part a.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.ii only.

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7 Chymosin is an enzyme found in young humans, but not in adults. It converts soluble proteins in milk into solid proteins.

(a) (i) Suggest the advantages to a young human of having chymosin in their alimentary canal.

(2)

(ii) Explain why chymosin is only needed in young humans.

(2)

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(iii) Describe a test to show that the solid formed by the action of chymosin on milk is a protein.

(3)

6. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

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